In [1]:
```
# Import Libraries

import pandas as pd
import numpy as np

from sklearn.linear_model import LinearRegression
import joblib    # for saving the model


# Load the dataset

data = pd.read_csv("Linear-Regression-_Training-Data_-.csv")

data.head(25)
```

Out[1]:

	Purchase hybrid vehicle	Age	Education	Income	Vehicle Ownership
0	0	4	2	4	1
1	0	1	2	9	1
2	0	1	3	7	2
3	1	3	4	12	4
4	0	2	3	7	2
5	0	1	2	7	2
6	0	2	3	9	1
7	0	1	1	6	1
8	0	1	2	5	2
9	0	1	2	7	2
10	0	2	1	7	1
11	0	2	2	5	2
12	0	2	3	5	1
13	1	2	2	10	3
14	0	1	3	7	2
15	0	1	4	7	1
16	0	1	1	9	1
17	0	2	4	5	1
18	0	1	3	7	1
19	0	2	2	2	1
20	0	1	3	6	1
21	0	3	2	5	1
22	1	2	3	12	3
23	0	2	2	10	1
24	0	1	2	6	1

In [2]:
```
# Define the Independent Variables (X) and Dependent Variables (y)

X = data[["Age","Education","Income","Vehicle Ownership"]]
y = data["Purchase hybrid vehicle"]

# Initialise & train the linear regression model

model = LinearRegression()
model.fit(X,y)

# Save the trained model to a file for later use

joblib.dump(model,"hybrid_vehicle_prediction_model.pkl")

print("Model has been trained and saved successfully")
```

```
Model has been trained and saved successfully
```

In [11]:
```
# Load the test dataset

new_data = pd.read_csv("Linear-Regression-Python-_Test-data_.csv")

new_data.head(20)
```

Out[11]:

	Age	Education	Income	Vehicle Ownership
0	3	2	8	2
1	4	4	12	3
2	2	3	10	1
3	5	5	15	4
4	1	2	5	1
5	3	3	9	2
6	2	1	6	1
7	4	3	11	3
8	3	2	8	2
9	5	4	13	4
10	2	3	7	2
11	1	1	4	1
12	3	2	9	2
13	4	4	12	3
14	5	3	11	4
15	2	2	6	2
16	1	3	7	1
17	4	2	10	3
18	3	4	12	2
19	5	5	14	4

In [12]:
```
# Load the trained model from the saved file

ml_model = joblib.load("hybrid_vehicle_prediction_model.pkl")


#  Make predictions using the model on the new input data (new_data)

new_data["Prediction %"] = ((ml_model.predict(new_data))*100).round(2)

new_data.head(20)
```

Out[12]:

	Age	Education	Income	Vehicle Ownership	Prediction %
0	3	2	8	2	47.80
1	4	4	12	3	108.53
2	2	3	10	1	48.74
3	5	5	15	4	156.74
4	1	2	5	1	-3.35
5	3	3	9	2	60.32
6	2	1	6	1	7.63
7	4	3	11	3	96.01
8	3	2	8	2	47.80
9	5	4	13	4	136.18
10	2	3	7	2	36.82
11	1	1	4	1	-15.87
12	3	2	9	2	55.84
13	4	4	12	3	108.53
14	5	3	11	4	115.63
15	2	2	6	2	24.30
16	1	3	7	1	17.20
17	4	2	10	3	83.49
18	3	4	12	2	88.92
19	5	5	14	4	148.70

In [14]:

```
# Download the csv file

new_data.to_excel("Samraat_Prediction.xlsx", index=False)
```

In [ ]: